

# The Marchetti Settlement

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The obligation was exact: \$50,000,000, payable on 15 March 2034, to a trust administering a structured settlement. There was no flexibility in the amount and none in the date.

Rosalind Achebe managed the reserve portfolio that stood behind it at Carrick Mutual, and had inherited the position in December. Her predecessor had funded the obligation by buying a portfolio of high-grade corporate bonds selected on yield, with maturities scattered between two and nineteen years. The portfolio's market value was \$36.0 million against a present value of the liability of \$36.0 million at the prevailing 4.2 percent flat curve, so on the day Achebe took over, the position was exactly funded.

Her predecessor's file note explained the strategy in one line: buy the highest yields available in investment grade, hold to maturity, and the obligation takes care of itself.

Achebe's concern was that being funded today at 4.2 percent said nothing about being funded in 2034. If rates fell, the liability's present value would rise faster than the portfolio's, because the liability was a single payment eight years out and the portfolio's cash flows arrived on average sooner. If rates rose, the reverse. The predecessor's strategy had no view on this at all; it had a view on yield.

She proposed to rebuild the portfolio around two liquid corporate issues: a three-year note with a 4 percent annual coupon and a fifteen-year bond with a 5 percent annual coupon, both currently priced to a 4.2 percent yield.

The investment committee approved the trade with one condition, imposed by a committee member named Fenwick who had been in the industry a long time and had a specific worry: that Achebe would present the position as risk-free once it was done. Fenwick wanted to see, in writing, the circumstances under which the immunized portfolio would fail.

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## EXHIBIT 1 — THE LIABILITY AND THE TWO BONDS

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|----------------------------|--------------------------------|
| Liability                  | \$50.0m due in exactly 8 years |
| Current curve              | 4.2% flat                      |
| Present value of liability | \$35.9773m                     |

|                   | BOND A  | BOND B   |
|-------------------|---------|----------|
| Maturity          | 3 years | 15 years |
| Annual coupon     | 4.0%    | 5.0%     |
| Price at 4.2%     | 99.4471 | 108.7716 |
| Macaulay duration | 2.8858  | 11.0873  |

Assume annual coupons and annual compounding throughout.

## ASSIGNMENT

- 1 Compute the portfolio weights in Bond A and Bond B that match the duration of the liability, and the face amount of each bond to buy. Verify that the portfolio's present value equals the liability's.
- 2 Immunization requires two conditions, not one. State both and confirm your portfolio satisfies both.
- 3 Shock the curve by plus 100, plus 200, plus 400, and minus 400 basis points, in parallel. Report the surplus or deficit in each case as a percentage of the liability. Explain the sign of the result.
- 4 Now apply a non-parallel shift: the three-year rate rises 150 basis points, the fifteen-year rate rises 25, and the eight-year rate rises 85. Report the surplus or deficit. Then reverse the twist and report it again.
- 5 Write Fenwick's memo. Under what circumstances does this portfolio fail, and what would you do about each?